
```

% datele de intrare
t_nod = [1900, 1910, 1920, 1930, 1940, 1950, 1960, 1970, 1980, 1990, 2000,
2010];
y_nod = [75.995, 91.972, 105.711, 123.203, 131.669, 150.697, 179.323,
203.212, ...
226.505, 249.633, 281.422, 308.786];

% punctele de aproximare pb4
t_eval = [1975, 2018];

disp('--- aproximare populatia SUA---');

% forma clasica pb 1, 4
y_clasic = lagrange_clasic(t_nod, y_nod, t_eval);
fprintf('metoda clasica:\n');
fprintf('Populatia in %d: %.3f milioane\n', t_eval(1), y_clasic(1));
fprintf('Populatia in %d: %.3f milioane\n', t_eval(2), y_clasic(2));

%forma baricentrica pb 5, 4
w = ponderi_baricentrice(t_nod);
y_baricentric = lagrange_baricentric(t_nod, y_nod, t_eval, w);
fprintf('\nmetoda baricentrica:\n');
fprintf('Populatia in %d: %.3f milioane\n', t_eval(1), y_baricentric(1));
fprintf('Populatia in %d: %.3f milioane\n', t_eval(2), y_baricentric(2));

%reprezentarea grafica pb 3
t_grafic = linspace(1900, 2020, 500);
y_grafic = lagrange_baricentric(t_nod, y_nod, t_grafic, w);

figure;
plot(t_nod, y_nod, 'ko', 'MarkerFaceColor', 'k', 'MarkerSize', 6); hold on;
plot(t_grafic, y_grafic, 'b-', 'LineWidth', 1.5);
plot(t_eval, y_baricentric, 'r*', 'MarkerSize', 8, 'LineWidth', 1.5);
title('Interpolare Lagrange: populatia SUA (1900-2010)');
xlabel('anul'); ylabel('populatie (milioane)');
legend('date recensamant (f)', 'polinom Lagrange (L_m f)', 'aproximari
(1975, 2018)', 'Location', 'northwest');
grid on;

%polinoame fundamentale pb2

plot_polinoame_fundamentale(t_nod);

%pb 1
function y_eval = lagrange_clasic(x, y, x_eval)
    m = length(x);
    n_eval = length(x_eval);
    y_eval = zeros(size(x_eval));

```

```

    for i = 1:n_eval
        suma = 0;
        for k = 1:m

            L_k = 1;
            for j = 1:m
                if j ~= k
                    L_k = L_k * (x_eval(i) - x(j)) / (x(k) - x(j));
                end
            end
            suma = suma + L_k * y(k);
        end
        y_eval(i) = suma;
    end
end

%pb5 p[onderi
function w = ponderi_baricentrice(x)
    m = length(x);
    w = ones(1, m);
    for j = 1:m

        w(j) = prod(x(j) - x([1:j-1, j+1:m]));
    end
    w = 1 ./ w;
end

%pb 5 interpolare
function y_eval = lagrange_baricentric(x, y, x_eval, w)
    m = length(x);
    y_eval = zeros(size(x_eval));
    exact = zeros(size(x_eval));

    num = zeros(size(x_eval));
    den = zeros(size(x_eval));

    for j = 1:m
        xdiff = x_eval - x(j);
        temp = w(j) ./ xdiff;
        num = num + temp * y(j);
        den = den + temp;
        exact(xdiff == 0) = j;
    end

    y_eval = num ./ den;

    idx = find(exact);
    y_eval(idx) = y(exact(idx));
end

%pb2

```

```

function plot_polinoame_fundamentale(x)
    m = length(x);
    x_grafic = linspace(min(x), max(x), 500);

    figure;
    hold on;
    for k = 1:m
        L_k = ones(size(x_grafic));
        for j = 1:m
            if j ~= k
                L_k = L_k .* (x_grafic - x(j)) / (x(k) - x(j));
            end
        end
        plot(x_grafic, L_k, 'LineWidth', 1.2);
    end
    plot(x, zeros(size(x)), 'ko', 'MarkerFaceColor', 'k');
    title('problema 2: polinoamele fundamentale l_k(x)');
    xlabel('x (Anii)'); ylabel('l_k(x)');
    grid on;
end

```

--- aproximare populatia SUA---

metoda clasica:

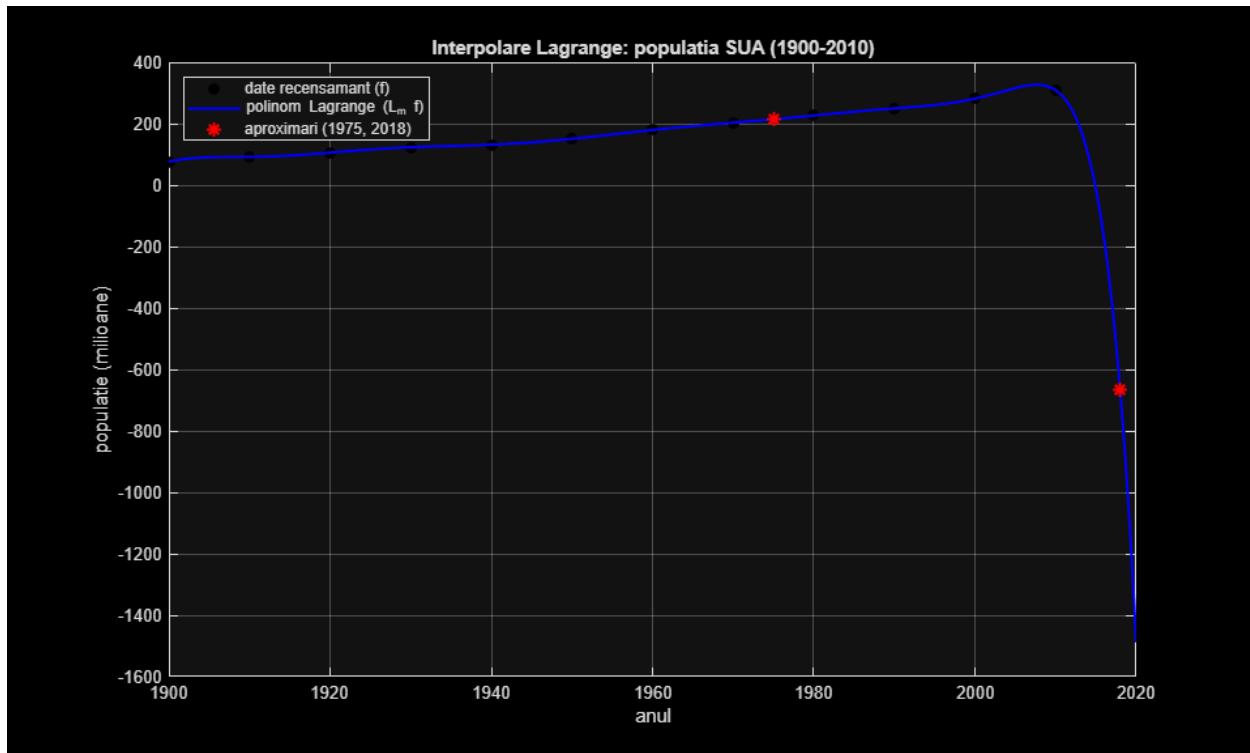
Populatia in 1975: 214.364 milioane

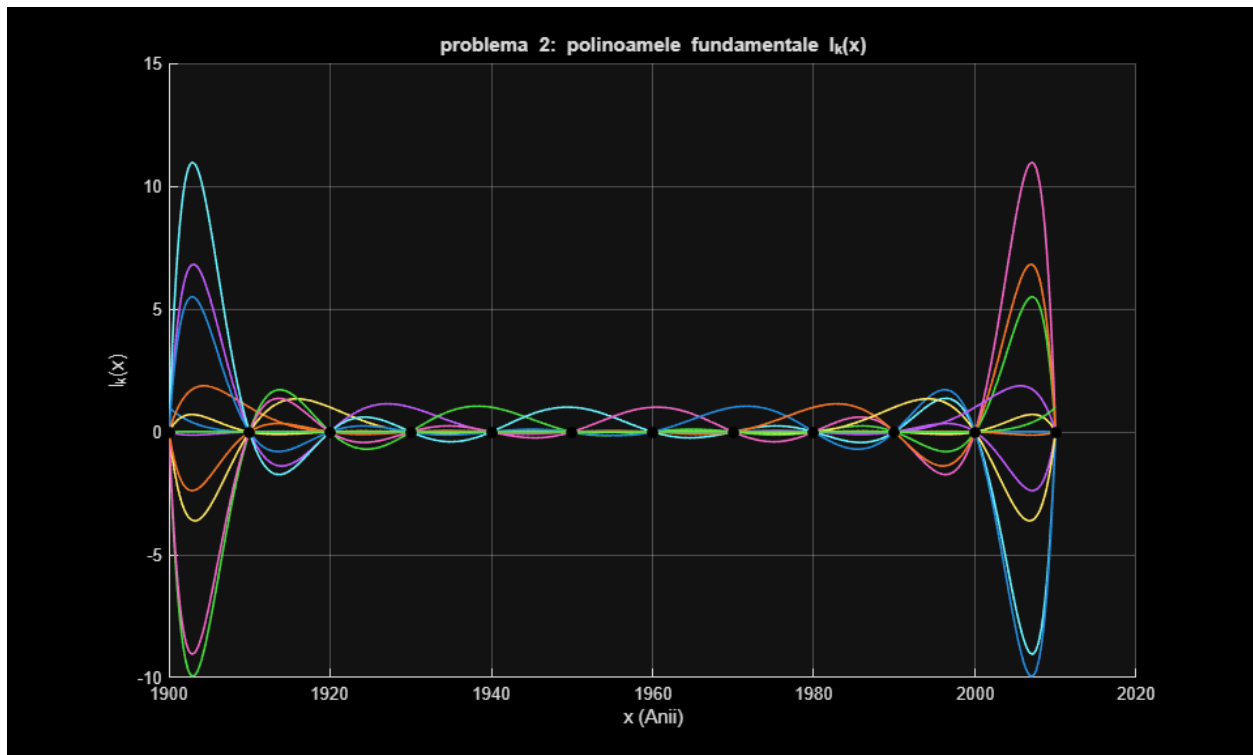
Populatia in 2018: -664.307 milioane

metoda baricentrica:

Populatia in 1975: 214.364 milioane

Populatia in 2018: -664.307 milioane





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